

Classification and risk assessment of ovarian-adnexal lesions using parametric and radiomic analysis of co-registered ultrasound-photoacoustic tomographic images.

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Ovarian-adnexal lesions are conventionally assessed with ultrasound (US) under the guidance of the Ovarian-Adnexal Reporting and Data System (O-RADS). However, the low specificity of O-RADS results in many unnecessary surgeries. Here, we use co-registered US and photoacoustic tomography (PAT) to improve the diagnostic accuracy of O-RADS. Physics-based parametric algorithms for US and PAT were developed to estimate the acoustic and photoacoustic properties of 93 ovarian lesions. Additionally, statistics-based radiomic algorithms were applied to quantify differences in the lesion texture on US-PAT images. A machine learning model (US-PAT KNN model) was developed based on an optimized subset of eight US and PAT imaging features to classify a lesion as either cancer, one of four subtypes of benign lesions, or a normal ovary. The model achieved an area under the receiver operating characteristic curve (AUC) of 0.969 and a balanced six-class classification accuracy of 86.0 %.

O-RADS MRI SCORE: An Essential First-Step Tool for the Characterization of Adnexal Masses.

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The ovarian-adnexal reporting and data system on magnetic resonance imaging (O-RADS MRI) score is now a well-established tool to characterize pelvic gynecological masses based on their likelihood of malignancy. The main added value of O-RADS MRI over O-RADS US is to correctly reclassify lesions that were considered suspicious on US as benign on MRI. The crucial issue when characterizing an adnexal mass is to determine the presence/absence of solid tissue and thus need to perform gadolinium injection. O-RADS MR score was built on a multivariate analysis and must be applied as a step-by-step analysis: 1) Is the mass an adnexal mass? 2) Is there an associated peritoneal carcinomatosis? 3) Is there any significant amount of fatty content? 4) Is there any wall enhancement? 5) Is there any internal enhancement? 6) When an internal enhancement is detected, does the internal enhancement correspond to solid tissue or not? 7) Is the solid tissue malignant? With its high value to distinguish benign from malignant adnexal masses and its high reproducibility, the O-RADS MRI score could be a valuable tool for timely referral of a patient to an expert center for the treatment of ovarian cancers. Finally, to make a precise diagnosis allowing optimal personalized treatment, the radiologist in gynecological imaging will combine the O-RADS MRI score with many other clinical, biological, and other MR criteria to suggest a pathological hypothesis. LEVEL OF EVIDENCE: 5 TECHNICAL EFFICACY STAGE: 3.

Gynecologic Adnexal Tumors and Tumor-Like Lesions in Children and Adolescents: Experience at a Cancer Center.

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Introduction. Tumors and tumor-like lesions of the uterine adnexa in children and adolescents are uncommon but may carry devastating consequences. **Methods.** We conducted an observational retrospective cohort study, to describe patients aged 0 to 19 years diagnosed with tumors and tumor-like lesions of the uterine adnexa at our institution between 2000 and 2018. **Results.** Eighty-nine patients

with 105 adnexal lesions were included. Thirty-seven (42%) patients presented with benign tumors, 13 (15%) with borderline tumors, 25 (28%) with malignant tumors and 14 (16%) with tumor-like lesions. Germ cell tumors ($n = 45|43\%$) were the most frequent, followed by epithelial tumors ($n = 30|29\%$). No significant differences were found in the age distribution of the lesions by malignant potential or histologic group. Most patients ($n = 80|90\%$) were treated primarily with conservative surgery, including cystectomy ($n = 25|28\%$) and unilateral oophorectomy/adnexectomy ($n = 54|61\%$). Thirty-four (38%) underwent surgical staging (partial omentectomy and peritoneal biopsies). Twenty (23%) patients with borderline and malignant tumors were submitted to chemotherapy. Four (5%) patients with borderline or malignant tumors relapsed, one of whom died from disease. Conclusion. Gynecological lesions in children and adolescents encompass a wide range of rare histological tumor subtypes, requiring evaluation by experienced pathologists. Most tumors were diagnosed at early stages, with low relapse and mortality rates. Conservative management, with fertility sparing surgery and limited use of adjuvant chemotherapy, is of utmost importance.

Magnetic resonance imaging evaluation of adnexal masses.

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Anatomically, the adnexa encompasses the region within the pelvis that includes the ovary, fallopian tube, round ligament, and structures arising from associated embryologic rests. The scope and spectrum of gynecologic pathology seen in this area is enormously diverse, and specific diagnoses are often elusive to conventionally-used imaging techniques such as ultrasound and CT. MRI has become an important tool in the evaluation of the adnexal mass because of its multiplanar capability and undersurpassed soft tissue contrast. It is particularly effective in defining the origin of a pelvic mass. The recent development of fast spin-echo sequences along with new phased array coils have enabled higher resolution imaging in shortened imaging times. The result is improved characterization of adnexal masses, which often leads to specific diagnoses.

Practice Bulletin No. 174 Summary: Evaluation and Management of Adnexal Masses.

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Adnexal masses (ie, masses of the ovary, fallopian tube, or surrounding tissues) commonly are encountered by obstetrician-gynecologists and often present diagnostic and management dilemmas. Most adnexal masses are detected incidentally on physical examination or at the time of pelvic imaging. Less commonly, a mass may present with symptoms of acute or intermittent pain. Management decisions often are influenced by the age and family history of the patient. Although most adnexal masses are benign, the main goal of the diagnostic evaluation is to exclude malignancy. The purpose of this document is to provide guidelines for the evaluation and management of adnexal masses in adolescents, pregnant women, and nonpregnant women and to outline criteria for the identification of adnexal masses that are likely to be malignant and may warrant referral to or consultation with a gynecologic oncologist.

Clinicopathological profile of adnexal masses presenting to a tertiary-care hospital in Bhutan.

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Adnexal masses of ovarian origin are of growing concern due to high fatality associated with ovarian malignancy because they are diagnosed at advanced stage due to vague symptoms and absence of recommended screening tests. The aim was to study the prevalence of histopathologic types of adnexal masses in different age groups and to analyze the accuracy of preoperative evaluation in diagnosing ovarian malignancy. This was a cross-sectional study carried out in the Department of Obstetrics and Gynecology of a tertiary care hospital in Bhutan with gynecologic-oncology services, from January to December 2017. Women presenting with adnexal mass were evaluated and those meeting criteria were enrolled. They were evaluated preoperatively with complete history, examination, ultrasound, and tumor markers. Risk of malignancy index (RMI) was calculated for all patients. Following surgery, histopathology results were compared with preoperative evaluation. Chi-square test, t-test, Cohen's Kappa, and receiver operating characteristic curve analysis were used for statistical analysis. Of 165 patients evaluated, 127 fulfilling criteria were enrolled. Adnexal masses of ovarian origin were most common ($n = 102$, 80.3%), of which 12.7% were malignant. Epithelial ovarian malignancy was the most common malignant ovarian tumor, serous cystadenocarcinoma being the most common. Malignancy was significantly more in older, postmenopausal women with high RMI. Seven out of 11 women with high RMI were diagnosed in Stage 3 or 4. RMI score at cutoff of 200 was 54.6% sensitive and 85.7% specific. Adnexal mass of ovarian origin was the most common. Malignancy was significantly more in older, postmenopausal women with high RMI. RMI showed moderate correlation in diagnosing epithelial ovarian malignancies.

Ultrasound characteristics of different types of adnexal malignancies.

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To describe ultrasound characteristics of adnexal malignancies, i.e., borderline ovarian tumors, primary invasive ovarian epithelial cancer stage 1, primary invasive ovarian epithelial cancer stages 2-4, rare types of malignancy, and metastatic tumors. In a prospective international study involving nine European ultrasound centers, 1,066 women with a pelvic mass judged to be of adnexal origin underwent transvaginal gray scale and color Doppler ultrasound examination by a skilled examiner before surgery. A standardized examination technique and predefined definitions of ultrasound characteristics were used. Of 1,066 masses, 266 were malignant and are included: 55 ovarian borderline tumors, 144 primary invasive epithelial ovarian cancers (42 stage 1, 102 stages 2-4), 25 rare malignancies, and 42 metastatic tumors. Most (56%) metastatic tumors and most (60%) rare types of tumor were solid and richly vascularized at color Doppler ultrasound examination (on a scale ranging from 1 to 4, color score based on subjective evaluation was 3 or 4 in 88% and 86%, respectively). Borderline ovarian tumors and stage 1 primary invasive ovarian epithelial cancers differed from stages 2-4 primary invasive ovarian epithelial cancers: they were larger (median volume 375 ml and 695 ml vs. 209 ml; $P = 0.0213$ and 0.0001), a larger proportion contained papillary projections (64% and 67% vs. 41%; $P = 0.0072$ and 0.0054), they were more often multilocular cysts without solid components (18% and 14% vs. 2%; $P < 0.0017$ and 0.0204), but they were less often purely solid (5% and 7% vs. 38%; $P \leq 0.0001$ and 0.0005). With increasing degree of invasiveness - from borderline epithelial ovarian tumors via stage 1 invasive epithelial ovarian tumors to stages 2-4 invasive epithelial ovarian tumors - ascites became more common (9% vs. 31% vs. 61%; $P = 0.0082$, <0.0001 , and 0.0017), and, among tumors with solid components ($n = 179$), the proportion of tumor consisting of solid tissue increased (median 2%-10%-34%; $P = 0.0212$,

<0.0001, and 0.0003). Papillary projections are characteristic of borderline tumors and stage 1 primary invasive epithelial ovarian cancer. A small proportion of solid tissue at ultrasound examination makes a malignant mass more likely to be a borderline tumor or a stage 1 epithelial ovarian cancer than an advanced ovarian cancer, a metastasis, or a rare type of tumor.

Primary sources of pelvic serous cancer in patients with endometrial intraepithelial carcinoma.

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Serous endometrial intraepithelial carcinoma is often associated with extrauterine disease. It is currently unclear where does the extrauterine disease come from. This study addressed this issue. A total of 135 samples from 21 serous endometrial intraepithelial carcinoma patients were studied. Cellular lineage relationships between intrauterine and extrauterine serous carcinomas were determined by TP53-mutation analysis and correlated to the clinicopathologic features. There were three conditions contributing the extrauterine disease: metastasis from serous endometrial intraepithelial carcinoma (n=10) showed identical TP53 mutation between intrauterine lesions and extrauterine disease, cases of adnexal origin (n=5) had discordant TP53 mutations, and the mixed cellular origin cases (n=6) with both identical and discordant mutation status. Patients with extrauterine disease from serous endometrial intraepithelial carcinoma metastasis typically had small tumor masses (<2 cm) in extrauterine sites and without finding of serous tubal intraepithelial carcinoma, while extrauterine disease with adnexal or tubal origin commonly had larger tumor masses in extrauterine sites including ovary and omentum and serous tubal intraepithelial carcinoma. The majority of extrauterine diseases associated with serous endometrial intraepithelial carcinoma are metastasized from the endometrium. Serous endometrial intraepithelial carcinoma is frequently associated with serous cancers of adnexal or tubal origin, indicating that endometrial and adnexal or tubal serous cancers may share similar etiologies. TP53-mutation analysis provides a strong linkage for cellular lineage analysis. Tumor size in extrauterine disease and presence of serous tubal intraepithelial carcinoma or not are useful clinicopathologic features to determine primary cancer site, which helps in clinical management.

A practical approach to the ultrasound characterization of adnexal masses.

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Because pelvic ultrasound is commonly used to evaluate adnexal masses, it is important to understand the most useful sonographic features for predicting benign and malignant masses. Determining whether an adnexal mass is of ovarian or extraovarian origin is key in arriving at the most likely diagnosis. Most adnexal masses are benign, and each of the most common benign ovarian lesions has a typical sonographic appearance. Additionally, most malignant ovarian neoplasms have a solid component with detectable flow by Doppler ultrasound, allowing one to strongly suggest the diagnosis. We will review an approach to the ultrasound diagnosis of adnexal masses that progresses through a series of 4 questions to help lead one to the most likely diagnosis.

Magnetic resonance imaging of adnexal masses.

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Adnexal masses are common in women of all ages. A range of physiological and benign ovarian conditions that develop in women, especially in the reproductive age, and adnexal malignancies can be evaluated with magnetic resonance imaging (MRI). Management of women with adnexal masses is frequently guided by imaging findings; therefore, precise characterization of adnexal pathology should be performed whenever possible. Magnetic resonance imaging is useful in characterization of adnexal masses that are not completely evaluated by ultrasound because it can provide additional information on soft tissue composition of adnexal masses based on specific tissue relaxation times and allows multiplanar imaging at large field of view to define the origin and extent of pelvic pathology. The patients most likely to benefit from MRI are pregnant women and those who are premenopausal and have masses that have complex features on ultrasound but do not have raised cancer antigen 125 tumor marker levels. The overlap in imaging appearance among different cell type malignancies makes it difficult to predict exact histology based on MRI appearance; however, MRI has a high accuracy in differentiating benign from malignant masses. Teratomas, endometriomas, simple and hemorrhagic cysts, fibromas, exophytic or extrauterine fibroids, and hydrosalpinges can be diagnosed with high confidence. In this article, the authors review the histopathologic background and MRI features of adnexal masses and discuss the role of MRI in the differentiation of benign from malignant adnexal pathologies.
